MSE 718 Course Project - Group 6

University of Waterloo

Department of Management Science and Engineering

Bayesian Multilevel Analysis of Housing Price in Major Ontario Cities

Authors:
Henry A. Tetteh
Gary Zang
Aymen Mehrez

Date: March 28, 2026

Contents

1 Introduction **3**

2 Methods **4**
2.1 Data Construction 4
2.2 Model Specification 4
2.3 Priors and Estimation 5
2.4 Model Checks 6

3 Results **7**
3.1 Panel Overview 7
3.2 Main Posterior Estimates 7
3.3 City-Level Heterogeneity 8
3.4 Model Comparison 9
3.5 Posterior Predictive Checks 10

4 Discussion **11**
4.1 Model Comparison and Supply Effect 11
4.2 Heterogeneity Across CMAs 11
4.3 Limitations and Interpretation 11

1 Introduction

Housing affordability has become a major policy issue in Ontario as housing prices have risen sharply across many urban markets. In recent years, public debate has focused on several possible drivers of these price increases, especially changes in interest rates, growth in international student enrollment, and constraints in housing supply. These factors may affect cities differently because local housing markets vary in size, demand pressures, and construction activity. Understanding how these forces are related to housing prices is therefore important for evaluating current housing and immigration policy discussions.

This report examines how policy interest rates, international student enrollment, and housing supply are associated with housing prices in six major Ontario Census Metropolitan Areas (CMAs): Toronto, Ottawa-Gatineau, Hamilton, Kitchener-Cambridge-Waterloo, London, and St. Catharines-Niagara. Using an annual city-year panel from 2016 to 2023, the analysis models housing prices with a Bayesian multilevel approach that allows for both an overall effect across cities and city-specific variation. The goal is to assess whether student inflows and interest rates are meaningfully linked to housing prices, and whether controlling for housing supply changes those relationships.

2 Methods

2.1 Data Construction

The analysis uses a balanced annual panel for six Ontario CMAs: Toronto, Ottawa-Gatineau, Hamilton, Kitchener-Cambridge-Waterloo, London, and St. Catharines-Niagara. The final panel covers 2016 to 2023, giving 48 (6×8) city-year observations. Four local data sources were combined: the Statistics Canada New Housing Price Index (NHPI) [1], the Bank of Canada overnight policy rate [2], international student enrollment by institution [3], and CMHC housing starts [4]. The outcome variable is the log of the annual mean NHPI `Total (house and land)` series. Because the student data are available only at the academic-year level, all sources were harmonized to annual frequency. Monthly NHPI values were averaged within year, daily policy rates were averaged within year, and monthly housing starts were summed within year. International student counts were built by mapping Ontario institutions to the six target CMAs and summing enrollment within each CMA-year. For academic years like `2023/2024`, the earlier year is used for classification since it encompasses the majority of the study period. Institutions outside the six target CMAs were excluded. Suppressed values (`..`) were imputed as 0 for leading missing years (a university founded between 2016 and 2023 for example) and as the institution-specific mean for remaining missing values. After merging the four sources, the panel was complete with no missing values in the model variables.

2.2 Model Specification

The main analysis uses Bayesian multilevel Gaussian regression. Let y_{it} denote `log_price_index` for city i in year t. Two models were estimated. The baseline model is:

$$y_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2) \tag{1}$$

$$\mu_{it} = \beta_0 + u_{0i} + (\beta_1 + u_{1i})\text{Students}_{it}^{(s)} + \beta_2 \text{PolicyRate}_t \tag{2}$$

where $\text{Students}_{it}^{(s)}$ is standardized international student enrollment, PolicyRate_t is the annual mean policy rate, u_{0i} is a city-specific random intercept, and u_{1i} is a city-specific random slope for the international student-enrollment effect. The second model adds standardized housing starts as a supply control:

$$\mu_{it} = \beta_0 + u_{0i} + (\beta_1 + u_{1i})\text{Students}_{it}^{(s)} + (\beta_3 + u_{2i})\text{Supply}_{it}^{(s)} + \beta_2 \text{PolicyRate}_t \tag{3}$$

where $\text{Supply}_{it}^{(s)}$ is standardized housing starts and u_{2i} is a city-specific random slope for the housing supply effect. This setup allows each CMA to have its own baseline price level through the random intercept u_{0i}, capturing the fact that cities differ structurally in their price levels independent of the predictors. Beyond that, the average student effect

is estimated across all CMAs while still allowing that effect to differ by city through partial pooling on u_{1i}. In Model 2, the same partial pooling applies to the housing supply effect via u_{2i}, recognizing that the relationship between new construction and prices is not uniform across cities.

2.3 Priors and Estimation

Weakly informative priors were used throughout. The intercept was assigned `normal(4.8, 0.5)`, centered at the observed mean of `log_price_index` with a standard deviation deliberately wider than the entire data range (4.57–5.15), so the prior places essentially no constraint on the baseline price level and lets the likelihood dominate (the center 4.8 was taken from the empirical mean of the outcome, a mild form of empirical Bayes; the prior width is large enough relative to the data range that this has negligible influence on the posterior). The coefficient on standardized international student enrollment was assigned `skew_normal(0.1, 0.1, 3)`, reflecting the straightforward demand-side mechanism: international students require housing, which increases local demand and should exert upward pressure on prices. The skewed shape places more prior mass above zero than below, encoding that directional expectation while keeping the scale small enough (0.1) that the data can easily pull the estimate toward zero or negative if warranted. Coefficients for the policy rate and standardized housing supply were assigned `normal(0, 0.5)`, reflecting genuine uncertainty about direction for both variables. Group-level standard deviations were assigned `exponential(1)`, controlling how much cities are allowed to deviate from the population average — in their baseline price level and in their sensitivity to student enrollment and, in Model 2, to housing supply, which is also allowed to have a city-varying effect given that supply conditions differ structurally across CMAs. This prior produces the partial pooling that is the core benefit of the multilevel structure. The residual standard deviation received the same `exponential(1)` prior. The `lkj(2)` prior was placed on the correlation matrix among the random effects — a 2×2 structure in Model 1 (intercept and student slope) and a 3×3 structure in Model 2 (intercept, student slope, and supply slope) — expressing mild skepticism toward extreme correlations while allowing the data to reveal genuine relationships, such as whether cities with higher baseline prices also respond more strongly to student inflows. Both models were estimated in `brms` with the `rstan` backend using Hamiltonian Monte Carlo. Each model was fit with 4 chains, 2,000 iterations per chain, 1,000 warmup iterations, `adapt_delta = 0.99`, and `max_treedepth = 15`.

$$\beta_0 \sim \mathcal{N}(4.8,\ 0.5^2) \tag{4}$$

$$\beta_1 \sim \text{SkewNormal}(0.1,\ 0.1,\ 3) \tag{5}$$

$$\beta_2 \sim \mathcal{N}(0,\ 0.5^2) \tag{6}$$

$$\beta_3 \sim \mathcal{N}(0,\ 0.5^2) \qquad \text{(Model 2 only)} \tag{7}$$

For the random effects:

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim \mathcal{N}_2(\mathbf{0},\, \Sigma_1) \qquad \text{(Model 1)} \tag{8}$$

$$\begin{pmatrix} u_{0i} \\ u_{1i} \\ u_{2i} \end{pmatrix} \sim \mathcal{N}_3(\mathbf{0},\, \Sigma_2) \qquad \text{(Model 2)} \tag{9}$$

where the covariance matrix is decomposed as:

$$\Sigma = \mathrm{diag}(\boldsymbol{\tau})\, \mathbf{R}\, \mathrm{diag}(\boldsymbol{\tau}) \tag{10}$$

$$\tau_k \sim \mathrm{Exponential}(1), \qquad k = 1, \ldots, K, \quad K = \begin{cases} 2 & \text{Model 1} \\ 3 & \text{Model 2} \end{cases} \tag{11}$$

$$\mathbf{R} \sim \mathrm{LKJ}(2) \tag{12}$$

The residual standard deviation of the likelihood is assigned:

$$\sigma \sim \mathrm{Exponential}(1) \tag{13}$$

2.4 Model Checks

The Bayesian workflow included convergence checks, posterior predictive checks, out-of-sample model comparison, and an examination of city-specific heterogeneity. Convergence was assessed using $\hat{R}$ and bulk effective sample size (ESS) for all parameters, with thresholds of $\hat{R} < 1.01$ and bulk ESS > 400 (> 700 actually). Posterior predictive checking drew 500 simulated datasets from the posterior predictive distribution and summarized model fit through interval plots, density overlays by CMA, and empirical coverage rates for 50% and 95% predictive intervals. Predictive performance across models was compared using PSIS-LOO with moment matching, focusing on `elpd_loo`, `loo_compare()`, and Pareto-k diagnostics to identify influential observations. To justify the multilevel structure, a pooling comparison was conducted contrasting three estimation strategies: no pooling (city-by-city regressions), complete pooling (a single global model ignoring city identity), and partial pooling (the multilevel model). City-specific slopes for the student-enrollment effect were extracted from Model 1 to examine whether the effect is uniform across CMAs or concentrated in particular markets. Finally, an omitted variable bias assessment compared the student-enrollment coefficient between Model 1 and Model 2 to evaluate whether housing supply was acting as a confounder in the baseline specification.

3 Results

3.1 Panel Overview

The final dataset is a balanced annual panel of 48 observations for six Ontario CMAs from 2016 to 2023, with no missing values in the model variables. Descriptive patterns show that housing prices rose over most of the sample period. The average log(HPI) across cities increased from 4.58 in 2016 to a peak of 4.93 in 2022, before a slight retreat to 4.92 in 2023. Over the same period, total international student enrollment across the six CMAs grew from 91,905 to 218,241, with Toronto accounting for the majority of enrollment at an average of 78,116 students per year. The policy rate remained moderate through 2019 (1.75%), fell sharply during 2020–2021 to a low of 0.25%, and then rose steeply to 4.74% by 2023. Housing starts also increased over the panel, with construction activity heavily concentrated in Toronto (averaging 485 units per year versus fewer than 105 in any other CMA). Notably, international student enrollment and housing starts are highly correlated across the panel ($r = 0.95$), reflecting the tendency for cities that attract more students to also build more housing. This collinearity is relevant for interpreting the model coefficients, as separating the two effects within a small panel is inherently imprecise.

Table 1: Panel overview and summary statistics for the annual city-year dataset. The panel is balanced with 6 CMAs observed from 2016 to 2023, for a total of 48 observations.

Variable	*N*	Mean	SD	Min	Max
Price index	48	116.93	20.05	96.62	173.02
log(HPI)	48	4.749	0.157	4.571	5.153
Policy rate (%)	48	1.479	1.378	0.250	4.743
International students	48	24,132	26,050	3,708	105,729
Housing starts	48	125.91	165.99	22.54	571.05

Policy rate is a national variable; the same value applies to all six CMAs within a given year (8 unique annual observations).

3.2 Main Posterior Estimates

The Bayesian multilevel models show a positive association between international student enrollment and housing prices. In Model 1, the posterior mean for the standardized student-enrollment coefficient was 0.211, with a 95% credible interval of [0.088, 0.386], indicating that a one standard deviation increase in enrollment is associated with approximately a 21% increase in the housing price index on average across CMAs. The policy-rate coefficient was negative and credibly different from zero, with posterior mean −0.034 and a 95% credible interval of [−0.062, −0.003].

In Model 2, after adding standardized housing supply with a city-varying slope, the student-enrollment coefficient remained stable at 0.214 [0.092, 0.377], consistent with the omitted variable bias assessment which showed a shift of only 0.003. The policy-rate coefficient attenuated to −0.017 [−0.053, 0.027], with the credible interval now crossing zero, suggesting that once supply conditions are allowed to vary by city, the independent policy rate effect is no longer credibly distinguishable from zero. The housing-supply coefficient was positive but highly uncertain at 0.196 [−0.162, 0.661], providing no credible evidence of an independent supply effect at the population level.
The city-specific slopes from Model 1 reveal substantial heterogeneity in the student-enrollment effect across CMAs. Ottawa-Gatineau showed the strongest effect (1.556, 95% CI [1.048, 1.967]), followed by London (1.133 [0.660, 1.534]), Hamilton (0.855 [0.244, 1.433]), St. Catharines-Niagara (0.781 [0.343, 1.168]), and Kitchener-Cambridge-Waterloo (0.616 [0.420, 0.794]). Toronto showed the weakest effect (0.133 [0.022, 0.231]). All six credible intervals exclude zero, indicating a positive student-enrollment effect in every CMA, but the magnitude varies nearly

Table 2: Main posterior estimates from the two Bayesian multilevel models. Model 1 is the baseline specification, and Model 2 adds standardized housing supply.

Parameter	Model 1 Mean	Model 1 95% CI	Model 2 Mean	Model 2 95% CI
Intercept	4.8982	[4.4884, 5.3305]	4.9343	[4.5186, 5.3431]
International students (std.)	0.2111	[0.0877, 0.3862]	0.2141	[0.092, 0.3765]
Policy rate	-0.0343	[-0.0616, -0.0026]	-0.0166	[-0.0527, 0.0271]
Housing starts (std.)	–	–	0.1958	[-0.1618, 0.6608]

3.3 City-Level Heterogeneity

The multilevel structure reveals substantial heterogeneity in the student-enrollment effect across CMAs. All six city-specific slopes exclude zero, confirming a positive association in every market, but the magnitude varies nearly twelvefold. Ottawa-Gatineau showed the strongest effect (1.556, 95% CI [1.048, 1.967]), followed by London (1.133 [0.660, 1.534]), Hamilton (0.855 [0.244, 1.433]), St. Catharines-Niagara (0.781 [0.343, 1.168]), and Kitchener-Cambridge-Waterloo (0.616 [0.420, 0.794]). Toronto, despite having the largest absolute enrollment (averaging 78,116 students per year), showed the weakest effect (0.133 [0.022, 0.231]). This suggests that Toronto's housing market is driven by a broader set of demand-side forces beyond student enrollment, diluting the student-specific signal relative to smaller CMAs where international students represent a larger share of housing demand.
The pooling comparison further justifies the multilevel approach. No-pooling estimates (city-by-city regressions) produce wide and unstable credible intervals given only 8 observations per city, while complete-pooling (a single global model) forces an identical

slope across all CMAs and masks the heterogeneity documented above. Partial pooling strikes a balance by shrinking city-specific estimates toward the population mean in proportion to within-city data quality, producing more stable estimates without discarding genuine cross-city variation.

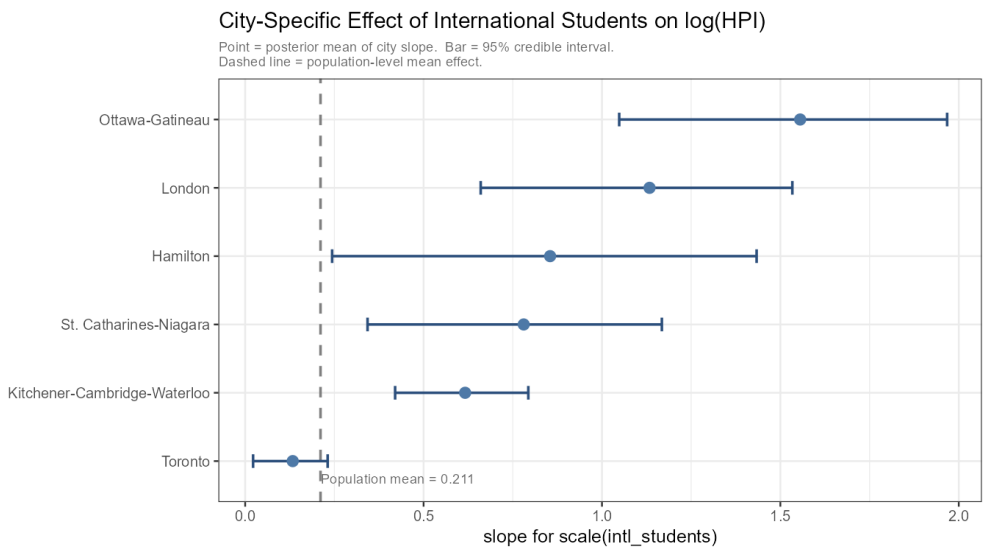

Figure 1: City-specific posterior slopes for the effect of international student enrollment on `log(HPI)` in Model 1. Points are posterior means, horizontal bars are 95% credible intervals, and the dashed line marks the population-level mean effect.

3.4 Model Comparison

Predictive performance was compared using PSIS-LOO with moment matching. Table 3 summarises the results. Model 1 achieved a higher expected log predictive density ($\widehat{\text{elpd}}_{\text{LOO}} = 53.4$, SE = 5.8) than Model 2 ($\widehat{\text{elpd}}_{\text{LOO}} = 49.4$, SE = 5.9). The difference is $\Delta\text{elpd} = -4.0$ (SE = 2.3), yielding a ratio of $|-4.0|/2.3 = 1.74$, which is below the conventional threshold of 2. The improvement in predictive accuracy from adding housing supply with a city-varying slope is therefore not meaningful. Model 2 also uses more effective parameters ($\hat{p}_{\text{LOO}} = 14.7$ vs 11.7), indicating that the added complexity of the supply random slope does not translate into better out-of-sample predictions.
Pareto-k diagnostics flagged 2 observations in Model 1 and 4 in Model 2 with $k > 0.7$, indicating that those specific city-years are too influential for the PSIS approximation to be fully reliable. These are expected given the small panel size of 48 observations and do not materially affect the comparison conclusion. Figure 2 shows the pointwise ELPD by CMA for Model 1. Predictive accuracy is broadly consistent across cities, with outlier years reflecting structurally unusual periods that no smooth trend could anticipate.

The omitted variable bias assessment confirms that the student-enrollment coefficient is stable across models (0.211 in Model 1 vs 0.214 in Model 2, shift = 0.003), providing no evidence that housing supply was confounding the student-enrollment effect in the baseline specification. Model 1 is therefore the preferred model.

3.5 Posterior Predictive Checks

Posterior predictive checks assessed whether the fitted models generate data consistent with the observed log(HPI) values. Five hundred datasets were simulated from the posterior predictive distribution and summarised through interval plots and empirical coverage rates.

Table 4 reports the empirical coverage of the 50% and 95% posterior predictive intervals by CMA. For the 95% interval, both models achieve 97.9% overall coverage (47 of 48 observations), with Kitchener-Cambridge-Waterloo as the only CMA falling below 100% (87.5% in both models). This is well-calibrated relative to the nominal target and confirms that the models do not systematically miss observed values. The 50% interval shows more variation across cities, with Hamilton exhibiting over-coverage and Ottawa-Gatineau exhibiting under-coverage in both models, reflecting that Ottawa-Gatineau's price trajectory is more variable than the model's predictive bands anticipate. With only 8 observations per CMA, city-level 50% coverage is inherently noisy and should be interpreted with caution.

Figure **??** shows the posterior predictive intervals over time for each CMA under Model 1. Observed values fall within the 95% bands for virtually all city-years, and the model captures the general upward trend across the panel. No systematic bias is evident for any single CMA, supporting the adequacy of the Gaussian likelihood assumption.

4 Discussion

4.1 Model Comparison and Supply Effect

PSIS-LOO favours Model 1 over the supply-adjusted Model 2 (Δelpd = -4.03, SE = 2.26, ratio = 1.78), below the conventional threshold of 2 for a meaningful difference. Adding housing supply with a city-varying slope increases model complexity without improving out-of-sample predictive accuracy, suggesting that supply does not carry additional predictive signal beyond what student enrollment and the policy rate already capture. The stability of the student-enrollment coefficient across models (shift = 0.003) further confirms that housing supply was not confounding the student-enrollment effect in Model 1. Both models reproduce the observed data well, with 95% posterior predictive interval coverage of 97.9%.

4.2 Heterogeneity Across CMAs

The nearly twelvefold variation in city-specific slopes (0.133 in Toronto to 1.556 in Ottawa-Gatineau) reflects differences in local market structure rather than estimation noise, since all six credible intervals exclude zero. In smaller CMAs, international students represent a larger share of total housing demand, so enrollment growth produces a more visible price response. In Toronto, student demand is a small signal within a much larger market driven by additional forces — speculative activity, foreign investment, and broader population growth — that dilute the enrollment effect.

This heterogeneity has a direct policy implication: a national international student cap would have unequal effects across cities, with smaller CMAs such as Ottawa-Gatineau and London bearing a disproportionately larger impact on housing demand than Toronto.

4.3 Limitations and Interpretation

The panel is small ($n = 48$, 8 years per CMA), limiting the precision of city-level estimates and giving disproportionate weight to any single unusual year. PSIS-LOO flagged 2 influential observations in Model 1 and 4 in Model 2 (Pareto-$k > 0.7$, none exceeding $k > 1$), consistent with this constraint.

The analysis operates at annual frequency, so within-year variation in housing prices and the policy rate is averaged out. The model also does not include a year fixed effect or explicit time trend, meaning some estimated associations may partly reflect shared temporal trends rather than distinct causal channels.

The high correlation between international student enrollment and housing starts ($r = 0.95$) means their independent effects cannot be cleanly separated. The student-enrollment coefficient should be read as the effect conditional on supply rather than an isolated demand effect. International student data is additionally aggregated from institution-

level counts using a manual CMA mapping, introducing potential measurement error for smaller cities.

The most defensible conclusion is therefore modest: international student enrollment and the policy rate show meaningful associations with housing prices across the six Ontario CMAs, while housing supply does not add clear predictive power in this small annual panel. These findings should be interpreted as associational rather than causal.

References

[1] Statistics Canada, "Statistics canada. table 18-10-0205-01 new housing price index, monthly," 2026. [Online]. Available: https://doi.org/10.25318/1810020501-eng pages 4

[2] Bank of Canada, "Interest rates and exchange rates," 2026, interest Rate (Bank of Canada). [Online]. Available: https://www.bankofcanada.ca/rates/interest-rates/canadian-interest-rates/ pages 4

[3] Statistics Canada, "Statistics canada. table 37-10-0232-01 postsecondary enrolments, by international standard classification of education, institution, and program and student characteristics," 2025. [Online]. Available: https://doi.org/10.25318/3710023201-eng pages 4

[4] ——, "Statistics canada. table 34-10-0156-01 canada mortgage and housing corporation, housing starts in all centres 10,000 and over, canada, provinces, and census metropolitan areas, seasonally adjusted at annual rates, monthly (x 1,000)," 2025. [Online]. Available: https://doi.org/10.25318/3410015601-eng pages 4

Appendix

Additional Tables

Table 3: PSIS-LOO model comparison. $\widehat{\text{elpd}}_{\text{LOO}}$ is the expected log predictive density; higher is better. $\hat{p}_{\text{LOO}}$ is the effective number of parameters. Δelpd is the difference relative to the preferred model (Model 1); a ratio $|\Delta\text{elpd}|/\text{SE} > 2$ is the conventional threshold for a meaningful difference.

Model	$\widehat{\text{elpd}}_{\text{LOO}}$	SE	$\hat{p}_{\text{LOO}}$	SE	Δelpd	SE
Model 1 (Base)	53.4	5.8	11.7	2.4	—	—
Model 2 (Supply-Adjusted)	49.4	5.9	14.7	2.7	−4.0	2.3

Table 4: Empirical coverage of the 50% and 95% posterior predictive intervals by CMA. Expected coverage under a well-calibrated model is 50% and 95% respectively. Each CMA has 8 observations (2016–2023).

	Model 1 (Base)		Model 2 (Supply-Adjusted)	
CMA	50%	95%	50%	95%
Hamilton	100.0	100.0	100.0	100.0
Kitchener-Cambridge-Waterloo	50.0	87.5	62.5	87.5
London	62.5	100.0	75.0	100.0
Ottawa-Gatineau	37.5	100.0	25.0	100.0
St. Catharines-Niagara	75.0	100.0	75.0	100.0
Toronto	87.5	100.0	100.0	100.0
Overall	68.8	97.9	72.9	97.9

Additional Figures

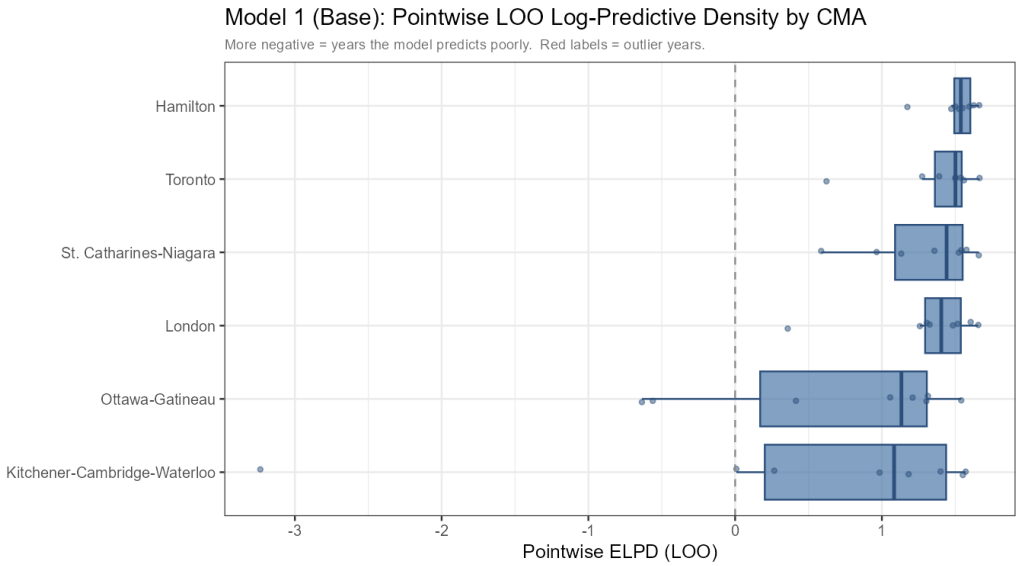

Figure 2: Pointwise LOO log-predictive density by CMA for Model 1. Each box covers the middle 50% of ELPD scores across the 8 years for that city. More negative values indicate years the model predicted poorly. Red labels identify outlier city-years.

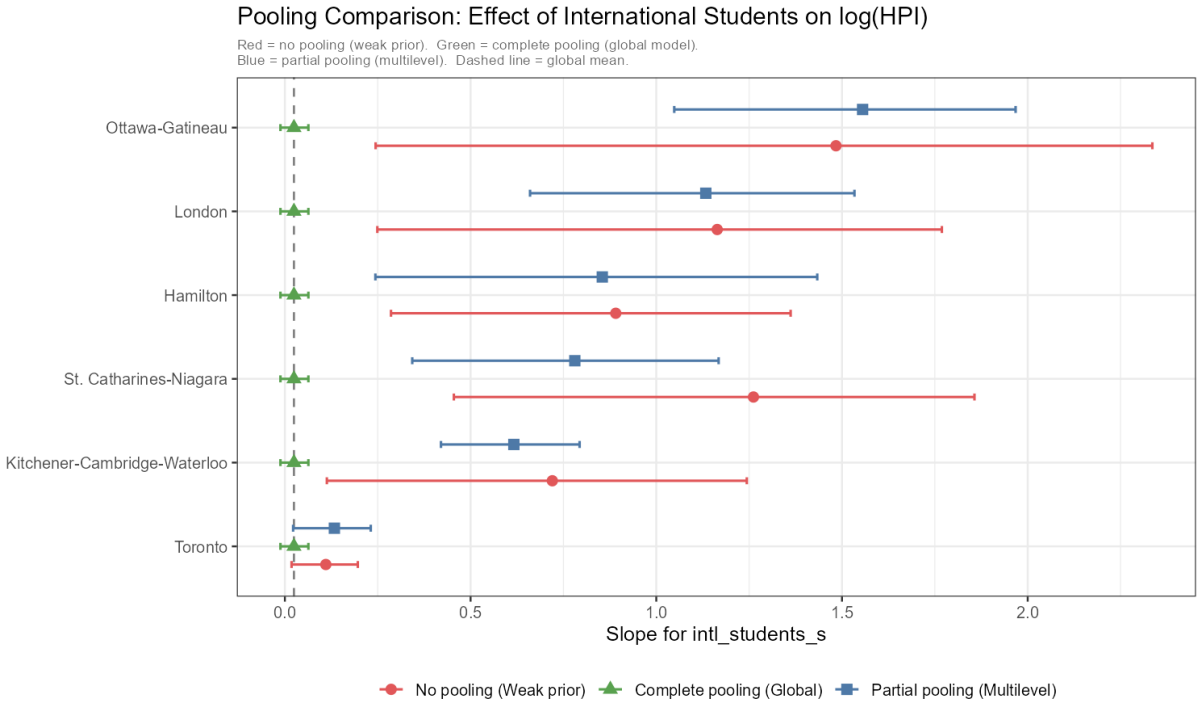

Figure 3: Comparison of no pooling, complete pooling, and partial pooling estimates for the international-student effect. The multilevel specification shrinks city-specific estimates toward the global mean while still preserving cross-city heterogeneity.

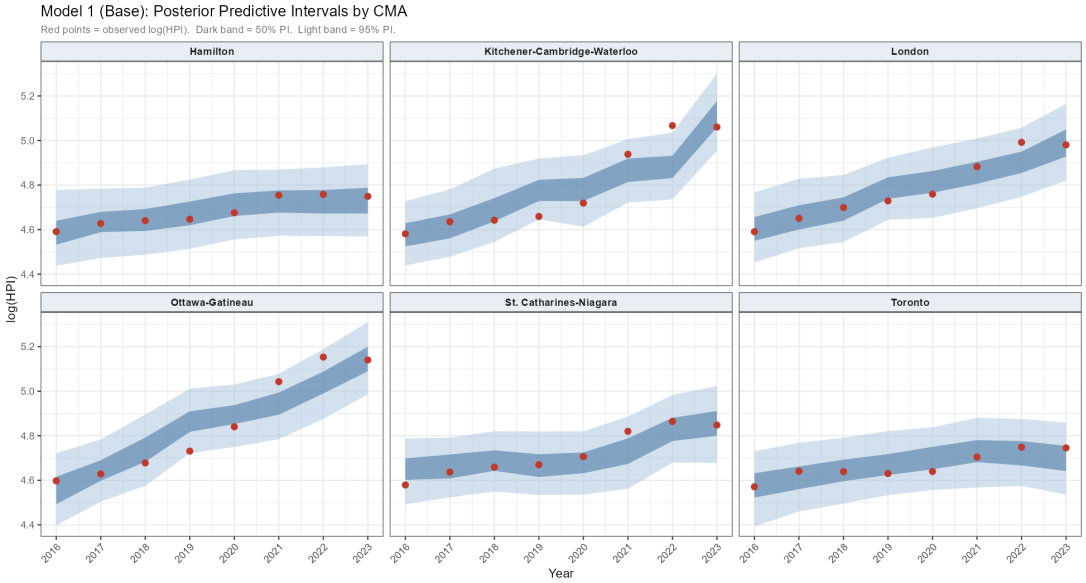

Figure 4: Posterior predictive intervals by CMA for Model 1. Red points are observed values of `log(HPI)`, while the dark and light bands denote the 50% and 95% posterior predictive intervals. Observed values falling outside the light band indicate years the model predicted poorly.

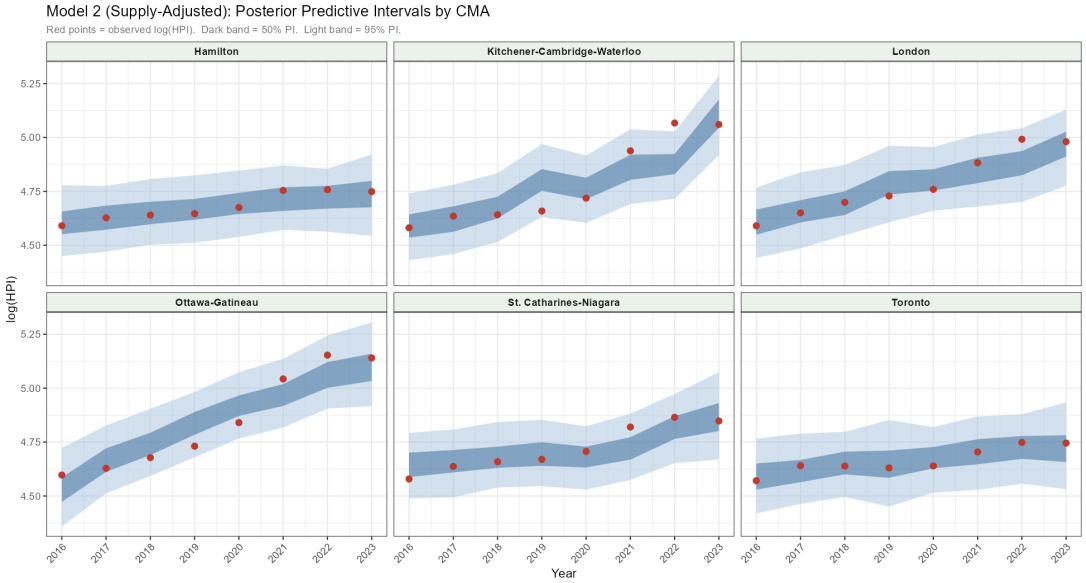

Figure 5: Posterior predictive intervals by CMA for Model 2, which adds housing supply and allows the housing-supply effect to vary across CMAs. Red points are observed values of `log(HPI)`, while the dark and light bands denote the 50% and 95% posterior predictive intervals. Model 2 shows weaker out-of-sample predictive performance than Model 1 in the PSIS-LOO comparison.

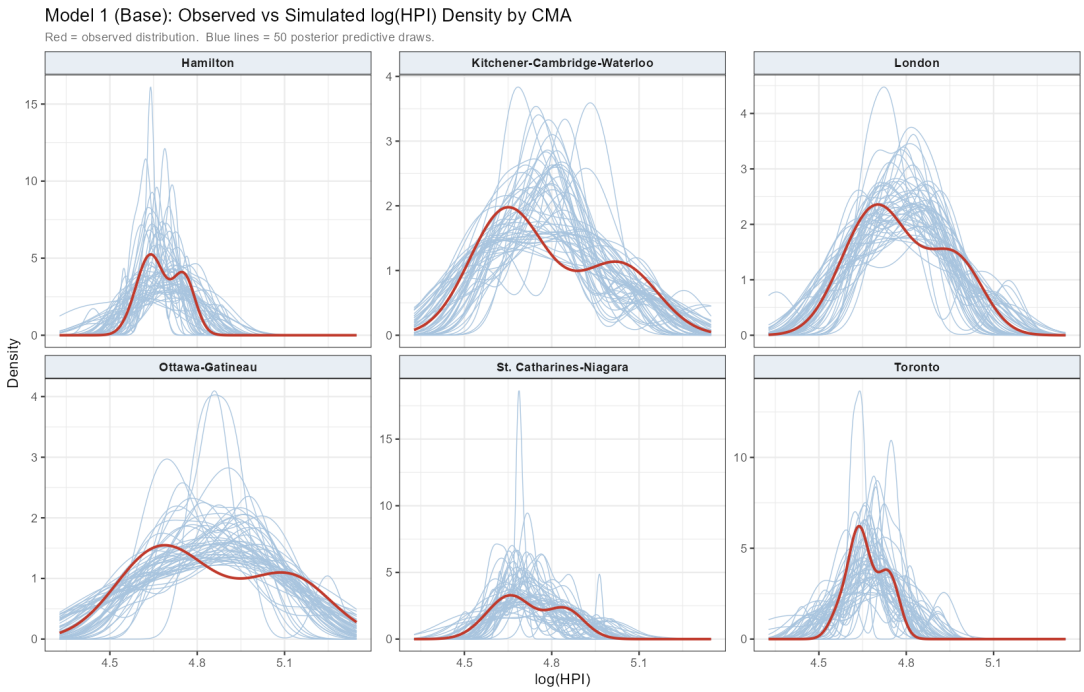

Figure 6: Observed and posterior predictive densities of `log(HPI)` by CMA for Model 1. The red curve is the observed distribution, and the blue curves are posterior predictive draws, showing that the model reproduces the overall distribution of housing prices reasonably well across cities.

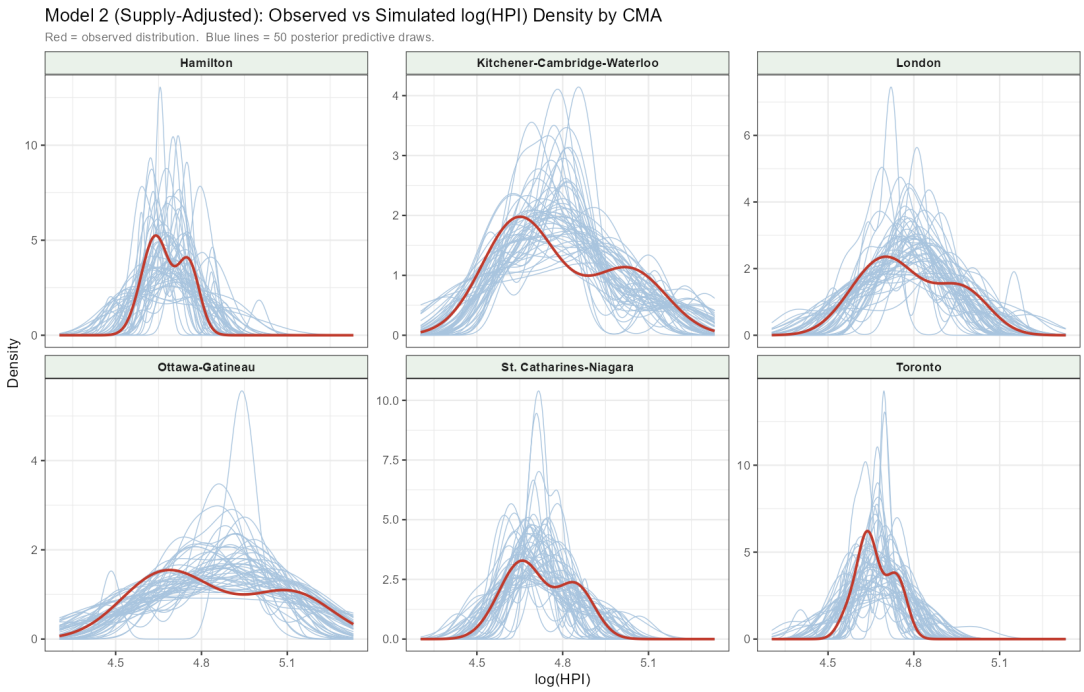

Figure 7: Observed and posterior predictive densities of `log(HPI)` by CMA for Model 2. The red curve is the observed distribution, and the blue curves are posterior predictive draws, showing that the model reproduces the overall distribution of housing prices reasonably well across cities.

AI Use Declaration

AI tools (including ChatGPT 5.4 and Codex) were used to assist with coding, data processing, and analytical implementation through terminal-based workflows. These tools contributed to statistical procedures such as data cleaning, modeling, and code execution. All AI-generated outputs were carefully reviewed, validated, and, where necessary, modified by the authors. The authors take full responsibility for the correctness, integrity, and interpretation of all analyses and results presented in this work. AI was also used for limited support in prose refinement, formatting, organizational tasks, and clarification of references. All uses of AI comply with the project's guidelines - use of AI tools.